

## **Final – MATH 518 – Winter 2025**

DATE: Tuesday, December 9      TIME: 2:00PM – 5:00PM

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- There are 6 problems, for a total of 70 points.
- Notes, calculators and other electronic devices are not permitted.
- Throughout the exam, the field  $k$  is of characteristic 0 and algebraically closed.
- The last pages can be used as scrap paper and will not be graded.

This page will not be graded and can be used as scrap paper.

1. (20 points) For each of the following statements, determine if it is true or false. Give a brief justification (short proof or counterexample).
  - (a) For certain integers  $n \geq 1$ , the quasi-projective variety  $X = k^n \setminus \{0\}$  is affine.
  - (b) The variety  $\mathbb{P}^1(k) \times \mathbb{P}^1(k)$  is neither affine nor projective.
  - (c) If  $X = V(f, g) \subseteq k^3$  is an affine variety given by two nonzero polynomials  $f, g \in k[x, y, z]$  of positive degree, then  $\dim(X) = 1$ .
  - (d) If an affine variety  $X$  is connected (in the Euclidean topology), then it is irreducible in the Zariski topology.
  - (e) Any two distinct lines in  $\mathbb{P}^2(k)$  intersect in a single point.

**Solution:**

- (a) TRUE. When  $n = 1$ , the open  $X = k \setminus \{0\}$  is affine, since the projection  $(x, y) \longrightarrow x$  induces an isomorphism between the hyperbola  $V(xy - 1)$  and  $X$ . On the other hand,  $k^2 \setminus \{0\}$  is not affine as we have seen in the homework.
- (b) FALSE. The variety  $\mathbb{P}^1(k) \times \mathbb{P}^1(k)$  is projective, since it can be embedded as a closed subvariety of  $\mathbb{P}^3(k)$  via the Segre embedding.
- (c) FALSE. If one takes  $f = g$ , then  $V(f)$  is of dimension 2.
- (d) FALSE. The variety  $X = V(xy) \subseteq k^2$  is the union of the  $x$ -axis and the  $y$ -axis. It is connected but reducible.
- (e) TRUE. A point  $[x : y : z] \in \mathbb{P}^2(k)$  is in the intersection of two lines  $a_1x + b_1y + c_1z = 0$  and  $a_2x + b_2y + c_2z = 0$  if and only if it is a solution of the corresponding  $2 \times 3$  system. Since the matrix has rank 2 (the lines being distinct), the set of solutions is 1-dimensional and gives a line in  $k^3$ , which corresponds to a single point in  $\mathbb{P}^2(k)$ .



2. (10 points) Find the irreducible components of

$$X = V(x^3 + yz, x(z + y), x^3 - z^2) \subseteq k^3.$$

Determine the dimension of each irreducible component.

**Solution:** From the equation in the middle we have  $x = 0$  or  $z = -y$ . This gives the decomposition

$$\begin{aligned} X &= V(x^3 + yz, x(z + y), x^3 - z^2) = V(x^3 - y^2, z + y) \cup V(yz, x, z^2) \\ &= V(x^3 - y^2, z + y) \cup V(x, z) \\ &= X_1 \cup X_2 \end{aligned}$$

3 decomp  
2 irred

since  $\text{rad}(yz, x, z^2) = (x, z)$ . For the first component  $X_1 = V(x^3 - y^2, z + y)$  we have

$$k[X] = k[x, y, z]/(x^3 - y^2, z + y) \simeq k[x, y]/(x^3 - y^2) = k[\bar{x}, \bar{y}]$$

induced by the map  $z \mapsto -y$ . The polynomial  $x^3 - y^2$  is irreducible in  $k[x, y]$ , so the coordinate ring is an integral domain and  $X$  is irreducible. The field of fractions is  $k(X_1) = k(\bar{x}, \bar{y})$ , which is algebraic over  $k(\bar{x})$ . Thus, the transcendence degree is one, and so  $\dim(X_2) = 1$ . 2.5

On the other hand,  $X_2$  is a line, so also of dimension 1. 2.5

In conclusion,  $X = X_1 \cup X_2$  is the union of two curves.



3. (10 points) Let  $f, g: X \rightarrow Y$  be morphisms between two quasi-projective varieties  $X$  and  $Y$ . If  $f$  and  $g$  agree on a nonempty open set  $U$ , show that they agree on  $X$ . *Show it for affine varieties first.*

**Solution:** Let us first show the statement for affine varieties  $X \subseteq k^n$  and  $Y \subseteq k^m$ . Let  $Z := \{p \in X \mid f(p) = g(p)\}$  be the set where  $f$  and  $g$  agree. Let us show that this is set closed. The morphisms are represented by two polynomial map  $f = (f_1, \dots, f_m)$  and  $g = (g_1, \dots, g_m)$  with  $f_i, g_i \in k[x_1, \dots, x_n]$ . Then  $p \in Z$ , if and only if  $f_i(p) - g_i(p) = 0$  for  $i \in \{1, \dots, m\}$ . It follows that  $Z = X \cap V(f_1 - g_1) \cap \dots \cap V(f_m - g_m)$  is closed in  $X$ . On the other hand, if  $f, g$  agree on  $U$ , then  $U \subseteq Z$ . But  $U$  is open so it is dense, and it follows that  $X = \overline{U} \subseteq Z \subseteq X$ . This shows that  $Z = X$ , and that  $f$  and  $g$  agree everywhere.

Let  $(U_\alpha)_{\alpha \in A}$  be a affine cover of  $X$  and  $(V_\alpha)_{\alpha \in A}$  a cover of  $Y$ , small enough that  $f(U_\alpha) \subseteq V_\alpha$ . If  $f$  and  $g$  agree on  $U$ , then  $f|_{U_\alpha}$  and  $g|_{U_\alpha}$  agree on each  $U \cap U_\alpha$ . For each  $\alpha \in A$  We have now two morphism  $f|_{U_\alpha}, g|_{U_\alpha}: U_\alpha \rightarrow V_\alpha$  of affine varieties, that agree on the nonempty open set  $U \cap U_\alpha \subseteq U_\alpha$ . We deduce from the first part of the proof that  $f|_{U_\alpha}$  and  $g|_{U_\alpha}$  agree on each affine piece  $U_\alpha$ , for all  $\alpha \in A$ . This shows that  $f, g$  agree everywhere on  $X$ .





4. (10 points) Consider the blowup map  $\phi: k^2 \longrightarrow k^2$  defined by  $\phi(x, y) = (x, xy)$ . Determine the image of  $\phi$  and show that it is a dense subset that is neither open nor closed.

2.5

**Solution:** The image of  $\phi$  is the set  $A := \{(0, 0)\} \cup U_x$  where  $U_x$  is the open subset where  $x \neq 0$ . The closure of  $U_x$  is  $k^2$ , so the closure of  $A$  is  $k^2$ . 2.5

To show that  $A$  is not open, we show that the complement  $k^2 \setminus A$  is not closed. The complement of  $A$  is  $k^2 \setminus A = V(y) \setminus \{0\}$ , the  $y$ -axis with the origin removed. A polynomial that vanishes on the dense open subset  $V(y) \setminus \{0\}$  of  $V(y)$ , must vanish on all of  $V(y)$  (by Q3 for example). It follows that  $k^2 \setminus A \subsetneq \overline{k^2 \setminus A} = V(y)$ . 2.5



5. (10 points) Let  $\phi: \mathbb{P}^1(k) \rightarrow \mathbb{P}^2(k)$  be the map defined by

$$\phi([u : v]) = [u^2 : u^2 - uv : v^2].$$

Let  $X = V(f)$  be the image of  $\phi$ . Determine  $f$  and show that  $\phi$  is an isomorphism onto  $X$ .

**Solution:** By restricting to the affine pieces  $U_x$  and  $U_y$ , we see that  $f$  is a morphism given by polynomial maps. Moreover, the image satisfies the equation  $f = (y - x)^2 - xz = 0$ . To show that  $\phi$  is an isomorphism onto  $X = V((y - x)^2 - xz)$  we define an inverse morphism  $\psi: X \rightarrow \mathbb{P}^2(k)$  on an open cover of  $X$ . If  $p \in X \cap U_y$ , then the equation  $(y - x)^2 - xz = 0$  shows that  $p \in X \cap U_x$  (otherwise  $y^2 = 0$ , contradicting  $y \neq 0$ ). So  $X$  is covered by  $X \cap U_x$  and  $X \cap U_z$ .

$$\psi([x : y : z]) = \begin{cases} [x : y - x] & \text{if } [x : y : z] \in X \cap U_x \\ [y - x : z] & \text{if } [x : y : z] \in X \cap U_z. \end{cases}$$

The map is a morphism on each piece, and the two agree on  $X \cap U_x \cap U_z$  since

$$[x : y - x] = [xz : (y - x)z] = [(y - x)^2 : (y - x)z] = [y - x : z].$$

(In the last step we use that  $y \neq x$ : if we had  $x = y$ , then  $xz = 0$  implies that  $x = 0$  or  $z = 0$ . Finally, we see that  $\psi$  and  $\phi$  are inverse to each other:

$$\psi \circ \phi([u : v]) = \psi([u^2 : u^2 - uv : v^2]) = \begin{cases} [u^2 : uv] = [u : v] & \text{if } u \neq 0 \\ [uv : v^2] = [u : v] & \text{if } v \neq 0. \end{cases}$$

The other direction is checked similarly.



6. (10 points) Let  $X \subseteq k^2$  be the curve  $X = V(f)$  where  $f = y^2 - x^2(1 + x)$ . Determine the points where  $X$  is smooth, and use blowups to find a smooth curve  $Y$  in  $k^2$  that is birational to  $X$ .

**Solution:** A point  $p \in X$  is singular if the derivatives of  $f$

$$\partial_x f = -2x - 3x^2 = -x(2 + 3x) = 0$$

$$\partial_y f = 2y = 0$$

both vanish. The points  $(-2/3, 0)$  is not in  $X$ :

$$f(-2/3, 0) = -(2/3)^2(1 - 2/3) \neq 0.$$

Thus, the curve  $X$  is smooth at every point except  $(0, 0)$ .

Let  $f(x, xy) = (xy)^2 - x^2 - x^3 = x^2(y^2 - 1 - x)$ . The blow-up at  $(0, 0)$  is the curve  $Y = V(f_1)$  where  $f_1(x, y) = y^2 - 1 - x$ , and the map  $\varphi: Y \rightarrow X$  defined by  $\varphi(x, y) = (x, xy)$  is birational. We have

$$\partial_x f_1 = -1 \neq 0$$

so  $Y$  is smooth curve that is birational to  $X$ .



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